Size selection (Purification) (1X ratio)

Huang MEGA Lab (NCKU)

(Lab plain-PDF edition, rebuilt 2026-10-20)

Source of record: Huang MEGA Lab (NCKU). Originally authored on protocols.io (dx.doi.org/10.17504/protocols. created 2026-09-13, version 1. This PDF is the lab's self-hosted copy and supersedes the protocols.io export.

Abstract

Size selection (Purification) with 1X ratio to keep more DNA than original version

Procedure

Procedure

---

1. Calculate the amount of magnetic beads you need and **DISTRIBUTE in a new 1.5 mL eppendorf tube (**Mix the bottle **VIGOROUSLY** to make sure it's homogeneous**).**

For example, if you have 5 samples to clean up, distribute 49.5 µl in a new 1.5 mL eppendorf tube.

Sample	Magnetic beads need
1	**25 µl**
2	**50 µl**
3	**75 µl**
4	**100 µl**
5	**125 µl**
6	**150 µl**
7	**175 µl**
8	**200 µl**

※ **Note:** The ratio is **25 µl of sample** use **25 µl of magnetic beads** to clean up (1 X to keep moderate to long DNA fragments).

※ **Note: Magnetic bead**: BeaverBeads™ DNA Select Isolation

---

3. (5 min) Mix sample and beads by **gently flicking** the tube for 5mins **5 min**

※ **Note:** Be gentle, mix too aggressive will make DNA fragmented

---

4. Transfer the tube to the 8-strip magnetic rack.

After all the magnetic beads are arrested, remove the supernatant.

---

5. Add **200 µL** 80% ethanol, flip the magnetic rack around to clean the pellet.

Wait until all the magnetic beads are arrested, remove the supernatant.

※ **Note: FRESHLY** make 80% ethanol

---

6. Repeat the **step 5**.

7. Flash spin the tube, and put on the magnetic rack.

Remove **ANY trace of liquid residuals** using 10 µl pipette.

※ **Note:** Caution: DO NOT let the beads crack

---

8. Add **15 µL** elution buffer.

Mix gently by flicking the tube and make sure all magnetic beads are dissolved in buffer, then flash spin down the sample.

---

10. Transfer the tube to the magnetic rack.

After all the magnetic beads are arrested, collect the **15 µL** supernatant to a 200 µl PCR tube or a 8-strip PCR tube.

---

11. The clean-up sample is now ready for 2' PCR.

---

9. (10 min) Incubate the tube in PCR thermal cycler using "**37_incubation**" program for 10min **10 min**

.

---

2. In each **25 µL** sample, add **25 µL** magnetic beads in.

※ **Note: Pipetting thoroughly every time** before adding to make sure all magnetic beads mixed well.